STPM 2019 PHYSICS 960/2 (U2)

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9$ N m^2 C^{-2}, $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}.

Q12: the copy available prints the option powers without minus signs; they are read as 10^{-2}, 10^{-3}, 10^{-4}.

Section A [15 marks]

1. The electric field on the surface of a positively charged conductor is

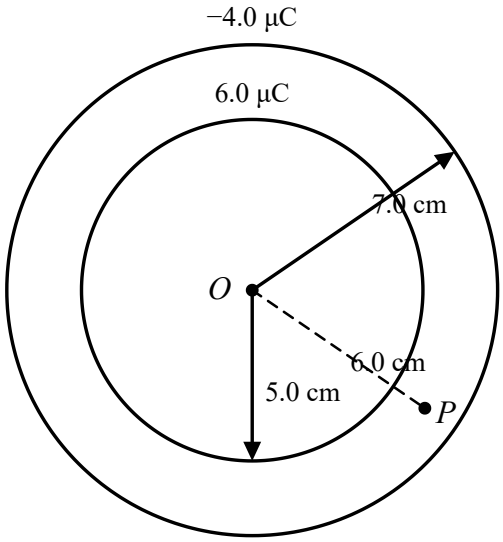
A always zero
B parallel to the surface
C perpendicular inwards to the surface
D perpendicular outwards to the surface

Answer: D

The surface of a conductor is an equipotential, so the field is perpendicular to it; for a positive charge it points outwards (D).

Explanation: Any component along the surface would move the charges.

2. Two concentric hollow metal spheres of radii 5.0 cm and 7.0 cm carry charges of 6.0 μC and −4.0 μC respectively is shown in the diagram below.



What is the electric field strength at *P* which is 6.0 cm from the centre *O*?

A 5.0×10^6 V m^{-1} B 7.7×10^6 V m^{-1} C 1.5×10^7 V m^{-1} D 2.2×10^7 V m^{-1}

Answer: C

At 6.0 cm only the inner sphere's charge (6.0 μC) is enclosed.

$$E = \frac{(9.0 \times 10^9)(6.0 \times 10^{-6})}{(0.060)^2} = 1.5 \times 10^7 \text{ V m}^{-1}$$

Explanation: The outer shell's charge produces no field inside it.

3. A capacitor consists of two parallel plates in which the distance between the plates can be adjusted. The capacitor is charged by connecting the capacitor to a battery, which is then removed. The potential difference between the plates of the isolated capacitor can be increased by

A touching the two plates
B bringing the two plates closer
C separating the two plates further apart
D allowing the potential difference increasing by itself

Answer: C

Isolated: Q constant; $V = \frac{Q}{C}$ and $C = \frac{\varepsilon_0 A}{d}$: increasing d decreases C and increases V

Explanation: Touching the plates discharges the capacitor.

4. A capacitor of capacitance 20 μF is discharged through a resistor of resistance 4.0 MΩ. After 25 s, the charge on the capacitor is 2.9×10^{-6} C. What is the time elapsed when the charge on the capacitor is 1.9×10^{-6} C?

A 59 s B 53 s C 40 s D 38 s

Answer: A

$$\tau = RC = (4.0 \times 10^6)(20 \times 10^{-6}) = 80 \text{ s}$$

$$\text{Extra time} = \tau \ln \frac{2.9}{1.9} = 80(0.423) = 34 \text{ s}; \; t = 25 + 34 = 59 \text{ s}$$

Explanation: Equal times give equal fractional decreases.

5. Two wires made of different materials have the same uniform current density. If both wires carry an equal amount of current, the wires can be deduced to have

A equal lengths
B equal cross-sectional areas
C equal potential differences
D equal lengths and cross-sectional areas

Answer: B

$J = \frac{I}{A}$: equal J and equal $I \Longrightarrow$ equal cross-sectional areas

Explanation: Nothing can be said about their lengths or p.d.s.

6. A uniform silicon wire of length 0.25 m is connected to a battery of potential difference 5.00 V. The electron density and electrical resistivity of the silicon wire are 1.45×10^{16} m^{-3} and 2.30×10^3 Ω m respectively. What is the drift velocity of the charge carriers of the silicon wire?

A 8.7×10^{-3} m s^{-1} B 5.0×10^{-2} m s^{-1} C 0.94 m s^{-1} D 3.7 m s^{-1}

Answer: D

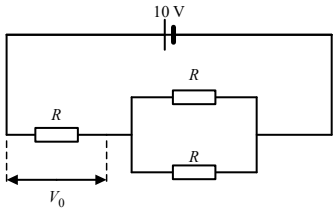
$$E = \frac{5.00}{0.25} = 20 \text{ V m}^{-1}; J = \frac{E}{\rho} = nev$$

$$v = \frac{20}{(2.30 \times 10^3)(1.45 \times 10^{16})(1.60 \times 10^{-19})} = 3.7 \text{ m s}^{-1}$$

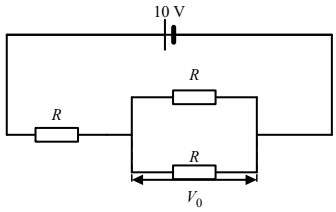
Explanation: The low carrier density of silicon gives a high drift velocity.

7. Identical resistors of resistance R are connected to an e.m.f. of 10 V. Which circuit has a potential difference, V_0, of 2.0 V?

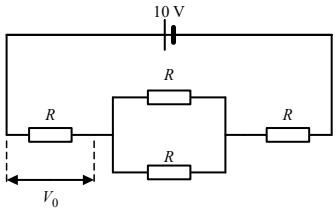
A



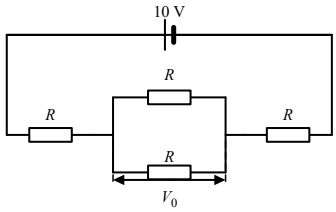
B



C



D

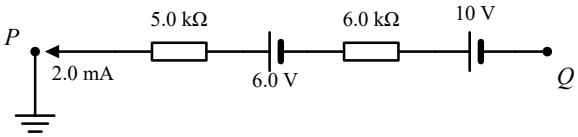


Answer: D

D: total resistance = $R + \frac{R}{2} + R = 2.5R$; the parallel pair has $\frac{0.5}{2.5}$ of the 10 V = 2.0 V.

Explanation: A gives 6.7 V, B gives 3.3 V, C gives 4.0 V.

8. A current of 2.0 mA flows from Q to P as shown in the diagram below.



If P is earthed, what is the electric potential at Q?

A 18 V B 12 V C 8.0 V D 4.0 V

Answer: A

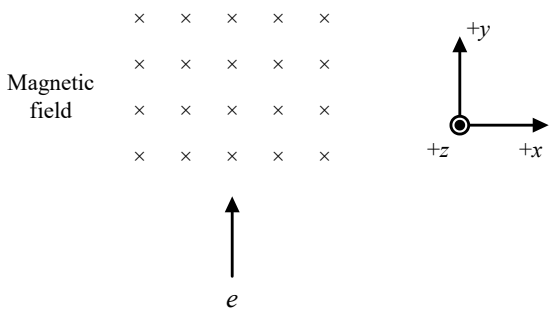
Going from P to Q against the current, the potential rises across each resistor: (2.0 mA)(5.0 kΩ) = 10 V and (2.0 mA)(6.0 kΩ) = 12 V.

The 6.0 V cell raises the potential by 6.0 V and the 10 V cell lowers it by 10 V (taking each cell's polarity as drawn):

$$V_Q = 0 + 10 + 6.0 + 12 - 10 = 18 \text{ V}$$

Explanation: Current flows from higher to lower potential through a resistor.

9. A region of uniform magnetic field in the $-z$ direction is shown in the diagram below.



An electron enters the magnetic field in the $+y$ direction. What is the direction of the magnetic force on the electron due to the magnetic field?

A $+x$ direction B $+y$ direction C $-x$ direction D $-y$ direction

Answer: A

$\mathbf{F} = -e(\mathbf{v} \times \mathbf{B})$; $\mathbf{v}$ along $+y$, $\mathbf{B}$ along $-z$: $\mathbf{v} \times \mathbf{B}$ is along $-x$, so $\mathbf{F}$ is along $+x$

Explanation: For an electron reverse the result of Fleming's left-hand rule.

10. A long thin straight wire carries a current which produces the magnetic flux density of 10 μT at a point 2.0 cm from the wire. If the current is doubled, what is the magnitude of the magnetic flux density at point 4.0 cm from the wire?

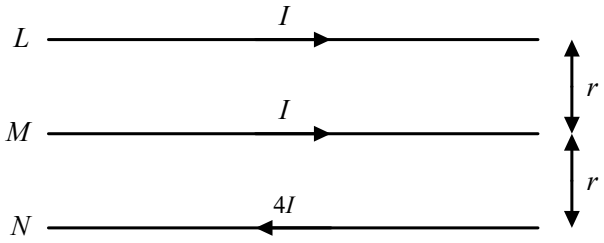
A 20 μT B 10 μT C 5.0 μT D 2.5 μT

Answer: B

$$B \propto \frac{I}{r}: B = 10\ \mu\text{T} \times \frac{2}{1} \times \frac{2.0}{4.0} = 10\ \mu\text{T}$$

Explanation: Doubling both the current and the distance leaves B unchanged.

11. Three long, thin and light conductors, L, M and N, carry currents, I, I and $4I$, respectively are placed parallel to each other at equal distance, r, apart as shown in the diagram below.



If the resultant force per unit length acts on L is F, what is the magnitude and direction of the resultant force per unit length acts on M?

	Magnitude	*Direction*
A	$3F$	Upward
B	$3F$	Downward
C	$5F$	Upward
D	$5F$	Downward

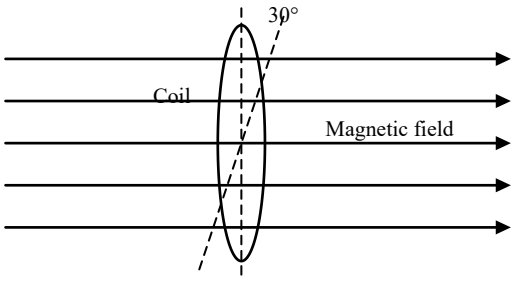
Answer: C

Let $k = \frac{\mu_0 I^2}{2\pi r}$. On L: k towards M (attraction), $2k$ away from N (repulsion): net k upward = F

On M: k upward (attracted to L) + $4k$ upward (repelled by N) = $5k = 5F$ upward

Explanation: Opposite currents repel; the force ∝ product of currents ÷ distance.

12. A coil placed in a horizontal uniform magnetic field of flux density 25 mT with its plane at an angle of 30° to the vertical as shown in the diagram below.



If the radius of the coil is 10.0 cm and the number of turns is 25, what is the magnetic flux through the coil?

A 2.0×10^{-2} Wb B 1.7×10^{-2} Wb C 9.8×10^{-3} Wb D 8.0×10^{-4} Wb

Answer: B

The plane is 30° from the vertical, so its normal makes 30° with the horizontal field.

$$N\Phi = NBA \cos 30° = (25)(25 \times 10^{-3})\pi(0.100)^2(0.866) = 1.7 \times 10^{-2} \text{ Wb}$$

Explanation: The answer is the total flux linking all 25 turns.

13. When current flows through an inductor of inductance 0.75 H, the energy stored in the inductor is 3.35 J. What is the current that flows through the inductor?

A 3.0 A B 4.5 A C 6.0 A D 12 A

Answer: A

$$I = \sqrt{\left(\frac{2U}{L}\right)} = \sqrt{\left(\frac{2(3.35)}{0.75}\right)} = 3.0 \text{ A}$$

Explanation: $U = \frac{1}{2}LI^2$.

14. A resistor of resistance 200 Ω is connected to a sinusoidal alternating current supply of peak voltage 340 V. Which statement is true about the circuit?

A The r.m.s. voltage of the circuit is 170 V.
B The r.m.s. current circuit is 1.70 A.
C The peak power dissipated by the resistor is 578 W.
D The average power dissipated by the resistor is 409 W.

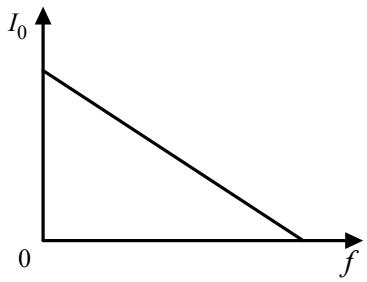
Answer: C

$$\text{Peak power} = \frac{V_0^2}{R} = \frac{340^2}{200} = 578 \text{ W}$$

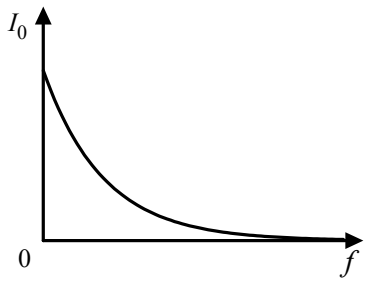
Explanation: $V_{rms} = 240$ V, $I_{rms} = 1.20$ A, average power = 289 W.

15. A resistor is connected in series to an inductor and an alternating current supply of constant peak voltage but of frequency, f, that can be varied. Which graph shows the variation of the peak current, I_0, in the circuit with the frequency, f?

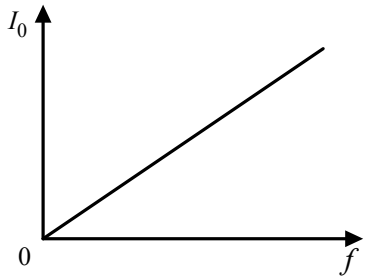
A



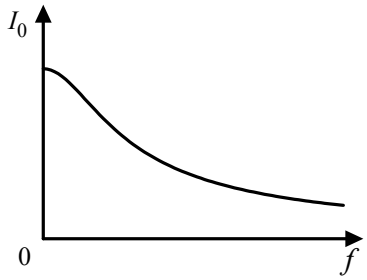
B



C



D



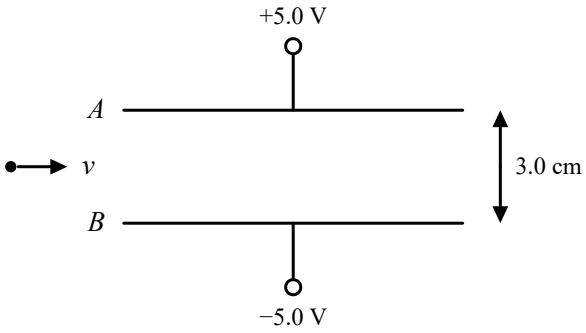
Answer: D

$I_0 = \dfrac{V_0}{\sqrt{(R^2 + (2\pi fL)^2)}}$: maximum V_0/R at $f = 0$, decreasing gradually towards zero (D)

Explanation: The curve starts flat because X_L is small compared with R at low frequency.

Section B [15 marks]

16. A point charge of +2.0 μC moves into a region of uniform electric field with a constant velocity, v, as shown in the diagram below.



The electric field is established between two conducting parallel plates, A and B, which are connected to a potential terminal of +5.0 V and −5.0 V respectively. The distance between the plates is 3.0 cm.

(a) Describe the path of the point charge which moves through the electric field. [3]

Answer

The charge experiences a constant force towards plate B (negative plate)	1
Its horizontal velocity stays constant while it accelerates uniformly towards B	1
So it follows a parabolic path bending towards B	1
	3

Explanation: Like a projectile under gravity.

(b) Calculate the electric field between the plates. [2]

Answer

$E = \dfrac{V}{d} = \dfrac{5.0 - (-5.0)}{0.030}$	1
$E = 333\ \text{V m}^{-1}$	1
	2

Explanation: The p.d. between the plates is 10 V.

(c) Determine the magnitude of the force experienced by the charge in traversing the electric field. [2]

Answer

$F = qE = (2.0 \times 10^{-6})(333)$ 1

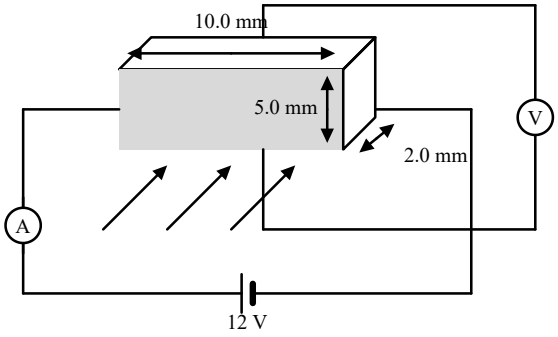
$F = 6.67 \times 10^{-4}$ N 1

2

Total for question **7**

Explanation: The force is the same everywhere in a uniform field.

17. An *n*-type semiconductor connected to a voltage supply of 12 V is placed perpendicularly in a uniform magnetic field and the semiconductor is connected to a voltmeter as shown in the diagram below.



The Hall voltage, current and the magnetic field are 4.0 mV, 0.026 A and 2.0 T respectively.

(a) Describe how the Hall voltage is produced. [3]

Answer

The electrons moving in the magnetic field experience a magnetic force *evB* perpendicular to their motion 1

They accumulate on one face, leaving the opposite face positive 1

This sets up a p.d. (Hall voltage) across the semiconductor, which grows until the electric force balances the magnetic force 1

3

Explanation: In equilibrium $eE_H = evB$.

(b) Calculate the drift velocity of the free electrons in the semiconductor. [3]

Answer

$eE_H = evB$ with $E_H = \frac{V_H}{d}$ 1

$v = \frac{V_H}{Bd} = \frac{4.0 \times 10^{-3}}{(2.0)(5.0 \times 10^{-3})}$ 1

$v = 0.40 \text{ m s}^{-1}$ 1

3

Explanation: d is the distance across which the Hall voltage is measured (5.0 mm).

(c) Determine the density of free electrons in the semiconductor. [2]

Answer

$n = \frac{I}{Aev} = \frac{0.026}{(5.0 \times 10^{-3})(2.0 \times 10^{-3})(1.60 \times 10^{-19})(0.40)}$ 1

$n = 4.1 \times 10^{22} \text{ m}^{-3}$ 1

2

Total for question **8**

Explanation: A is the cross-section through which the current flows.

Section C [30 marks]

18. (a) Describe the effect of inserting a dielectric material between the plates of the parallel plate capacitor. [3]

Answer

The dielectric is polarised; its induced surface charges reduce the electric field between the plates 1

For a given charge the p.d. is reduced 1

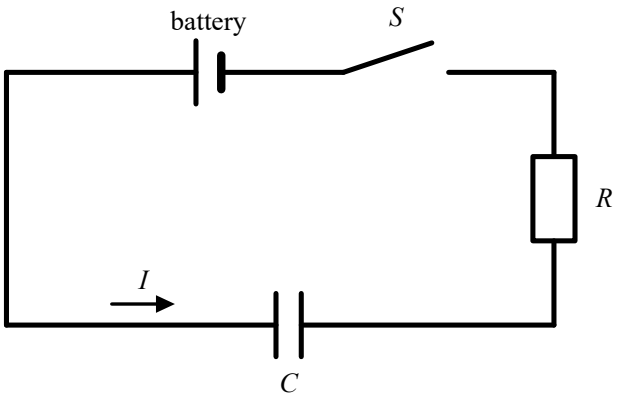
So the capacitance increases by a factor equal to the dielectric constant ε_r 1

3

Explanation: $C = \varepsilon_r C_0$.

(b) Describe the process of charging a parallel plate capacitor through a resistor with the aid of a suitable circuit. [4]

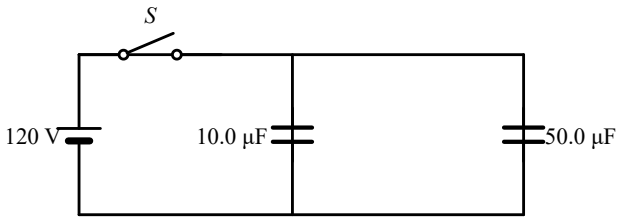
Answer



When S is closed, electrons flow from one plate through the battery to the other plate: the plates gain equal and opposite charges	1
The initial current is maximum, ε/R, as the capacitor is uncharged	1
As charge builds up, the p.d. across C increases and the p.d. across R (and the current) decreases exponentially	1
Charging stops when the p.d. across C equals the e.m.f.: $Q = Q_0(1 - e^{-t/RC})$	1
	4

Explanation: No charge crosses the gap between the plates.

(c) Two capacitors of capacitance 10.0 μF and 50.0 μF are connected in parallel and charged by a supply of 120 V when the switch, S, is closed as shown in the circuit below.



(i) S is then opened and both capacitors are allowed to discharge through a resistor of resistance 500 kΩ. Determine the time taken for the potential difference across both capacitors to decrease to 10.0 V. [4]

Answer

$C = 10.0 + 50.0 = 60.0\ \mu F$ 1

$\tau = RC = (500 \times 10^3)(60.0 \times 10^{-6}) = 30.0\ s$ 1

$V = V_0 e^{-t/RC} \Longrightarrow t = RC \ln \frac{V_0}{V} = 30.0 \ln \frac{120}{10.0}$ 1

$t = 74.5\ s$ 1

4

Explanation: Both capacitors discharge together through R.

(ii) If both capacitors are connected in series, calculate the required potential difference across both capacitors for the same amount of energy as stored in the parallel connection. [4]

Answer

Energy in parallel: $U = \frac{1}{2}(60.0 \times 10^{-6})(120)^2 = 0.432\ J$ 1

Series: $C = \frac{(10.0)(50.0)}{10.0 + 50.0} = 8.33\ \mu F$ 1

$V = \sqrt{(\frac{2U}{C})} = \sqrt{(\frac{2(0.432)}{8.33 \times 10^{-6}})}$ 1

$V = 322\ V$ 1

4

Total for question **15**

Explanation: A smaller capacitance needs a larger p.d. to store the same energy.

19. (a) An inductor of inductance 0.20 H with negligible resistance is connected in series with a resistor of resistance 9.0 Ω, a battery and a switch. The battery has an e.m.f. of 1.5 V and its internal resistance is 1.0 Ω.

(i) Explain the phenomenon of self-induction. [4]

Answer

A changing current in a coil produces a changing magnetic flux 1

This changing flux links the coil itself 1

An e.m.f. is induced in the same coil (Faraday's law) 1

By Lenz's law it opposes the change in current (a back e.m.f.): $\varepsilon = -L\ dI/dt$ 1

4

Explanation: Self-induction slows the growth and decay of current.

(ii) Calculate the rate of change of the current when the current is 0.10 A. [4]

Answer

$\varepsilon = I(R + r) + L\frac{dI}{dt}$ 1

$1.5 = (0.10)(9.0 + 1.0) + 0.20\frac{dI}{dt}$ 1

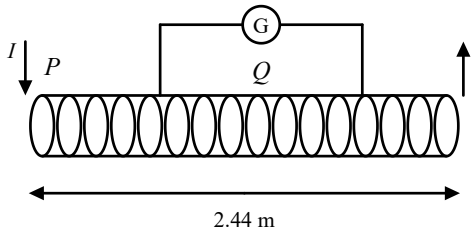
$\frac{dI}{dt} = \frac{1.5 - 1.0}{0.20}$ 1

$= 2.5\ \text{A s}^{-1}$ 1

4

Explanation: Include the internal resistance of the battery.

(b) A coil, Q, is wound on top of a long air-core coil, P, which carries a current, I, as shown in the diagram below.



P has a length of 2.44 m with 300 turns and cross-sectional area of 1.26×10^{-3} m^2 while Q has 100 turns. When the current in P changes, the pointer of the galvanometer connected to Q is deflected.

(i) Explain the phenomenon observed in Q. [4]

Answer

The changing current in P produces a changing magnetic flux 1

This changing flux links coil Q 1

An e.m.f. is induced in Q (mutual induction, Faraday's law), driving a current through the galvanometer 1

The induced current opposes the change in flux (Lenz's law); there is no deflection when the current in P is steady 1

4

Explanation: $\varepsilon_Q = -M\, dI_P/dt$.

(ii) If the current in P decreases uniformly from 12.0 A to zero in 58 ms, calculate the e.m.f. induced in Q. [3]

Answer

$M = \frac{\mu_0 N_P N_Q A}{l} = \frac{(4\pi \times 10^{-7})(300)(100)(1.26 \times 10^{-3})}{2.44} = 1.95 \times 10^{-5}$ H — 1

$\varepsilon = M\frac{\Delta I}{\Delta t} = (1.95 \times 10^{-5})\frac{12.0}{58 \times 10^{-3}}$ — 1

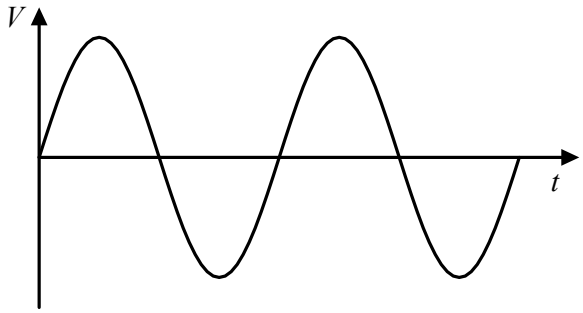
$\varepsilon = 4.0 \times 10^{-3}$ V — 1

3

Total for question **15**

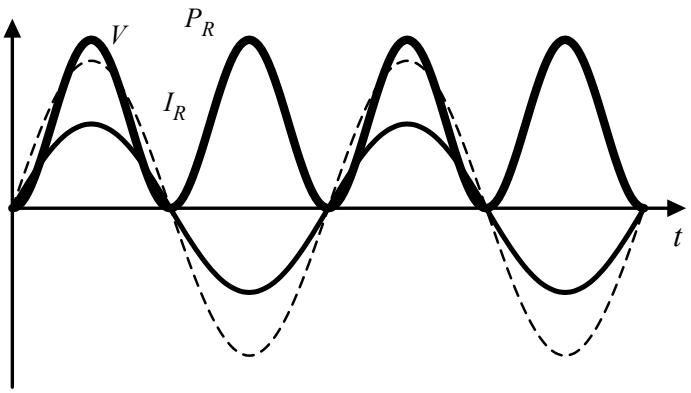
Explanation: All the flux of P passes through Q, which is wound on top of it.

20. (a) The waveform of a sinusoidal voltage which is supplied to a resistor is shown in the graph below.



(i) Sketch on the same axis of t, (1) the curve of the current, I_R, as a function of time, t; (2) the curve of the instantaneous power, P_R, as a function of time, t. [2]

Answer



I_R sinusoidal and in phase with V — 1

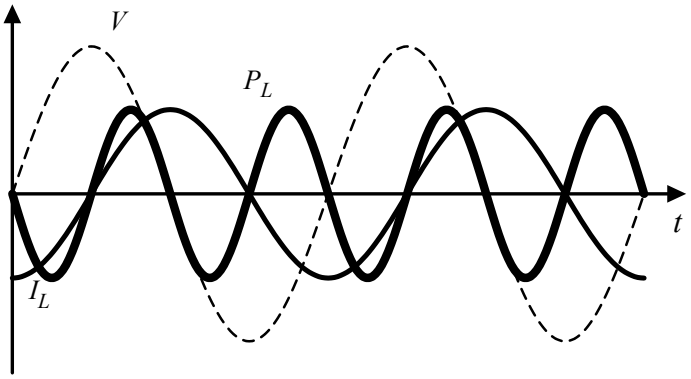
P_R always positive, with twice the frequency — 1

2

Explanation: $P = VI \propto \sin^2 \omega t$.

(ii) If the resistor is replaced by an inductor, sketch on the same axis of t on another graph, (1) the curve of the current, I_L, as a function of time, t; (2) the curve of the instantaneous power, P_L, as a function of time, t. [2]

Answer



I_L sinusoidal, lagging V by 90°	1
P_L alternately positive and negative, with twice the frequency (average zero)	1
	2

Explanation: $P \propto \sin \omega t \times (-\cos \omega t) = -\frac{1}{2} \sin 2\omega t$.

(b) A bulb labelled 120 V, 60 W is connected in series with a resistor, R, and an alternating current supply of r.m.s. voltage 230 V and frequency 50.0 Hz.

(i) Determine the value of R so that the bulb glows with normal brightness. [4]

Answer

Bulb current $I = \frac{P}{V} = \frac{60}{120} = 0.50$ A	1
P.d. across $R = 230 - 120 = 110$ V (in phase, so they add directly)	1
$R = \frac{110}{0.50}$	1
$R = 220\ \Omega$	1
	4

Explanation: The bulb is a resistor, in phase with R.

(ii) If R is replaced by an inductor, determine the self-inductance so that the bulb glows with the same normal brightness. [5]

Answer

$V^2 = V_{bulb}{}^2 + V_L{}^2$ (90° out of phase) 1

$V_L = \sqrt{(230^2 - 120^2)} = 196$ V 1

$X_L = \frac{196}{0.50} = 392\ \Omega$ 1

$L = \frac{X_L}{2\pi f} = \frac{392}{2\pi(50.0)}$ 1

$L = 1.25$ H 1

5

Explanation: Voltages across a resistor and an inductor add as phasors.

(c) State whether a resistor or an inductor is more suitable to be used to regulate the current in an alternating current circuit. Justify your answer. [2]

Answer

An inductor 1

Its average power dissipation is zero, so it limits the current without wasting energy as heat (a resistor would dissipate 55 W here) 1

2

Total for question **15**

Explanation: This is why chokes are used in a.c. circuits.